HeartSense

AI-DRIVEN CARDIAC INSIGHTS

Priyadharshini R
Abinaya T
Yuvaraj K

Overview

1) Exploratory Data Analysis

- Importing the required dataset from Kaggle
- Analysing different aspects of the data
- Attempting to find univariate and bivariate relations
- Making intuitive analyses that just make sense

2) Prediction

- Using the dataset to train different ML models
- Choosing the right model based on accuracy
- Predicting heart attack possibilities for new and unseen data

Data Dictionary

age	Age of the patient
sex	Sex of the patient
cp	Chest pain type: 0 = Typical Angina, 1 = Atypical Angina, 2 = Non-anginal Pain, 3 = Asymptomatic
trtbps	Resting blood pressure (in mm Hg)
chol	Cholestoral in mg/dl fetched via BMI sensor
fbs	Fasting blood sugar > 120 mg/dl ? 1 = True, 0 = False
restecg	Resting electrocardiographic results: 0 = Normal, 1 = ST-T wave normality, 2 = Left ventricular hypert
thalachh	Maximum heart rate achieved
oldpeak	Previous peak
slp	Slope
caa	Number of major vessels
thall	Thalium stress test result: (0, 3)
exng	Exercised induced angina: 1 = Yes, 0 = No
output	Target variable: 1 = Yes, 0 = No

Simple View of the Dataset

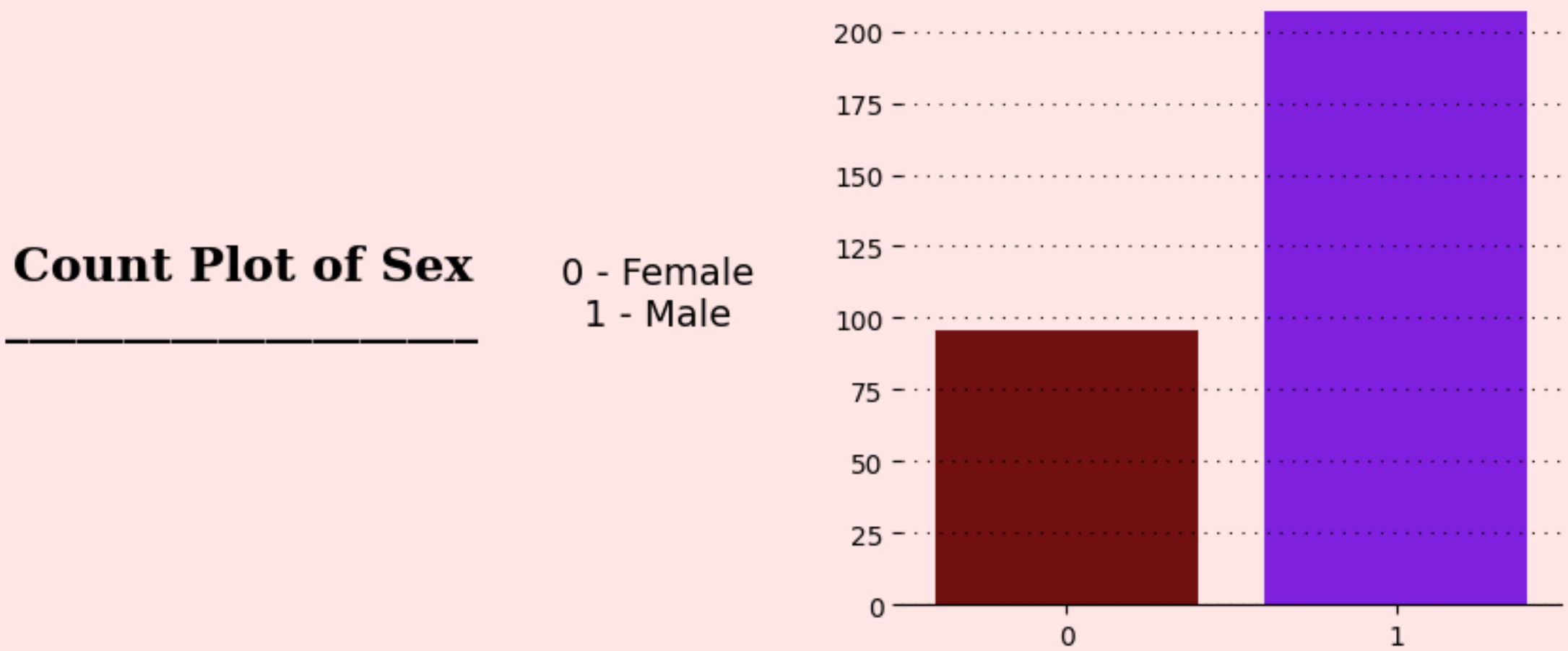
	age	sex	cp	trtbps	chol	fbs	restecg	thalachh	exng	oldpeak	slp	caa	thall	output
0	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
1	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
2	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
3	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
4	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
195	59	1	0	170	326	0	0	140	1	3.4	0	0	3	0
196	46	1	2	150	231	0	1	147	0	3.6	1	0	2	0
197	67	1	0	125	254	1	1	163	0	0.2	1	2	3	0
198	62	1	0	120	267	0	1	99	1	1.8	1	2	3	0
199	65	1	0	110	248	0	0	158	0	0.6	2	2	1	0

200 rows × 14 columns

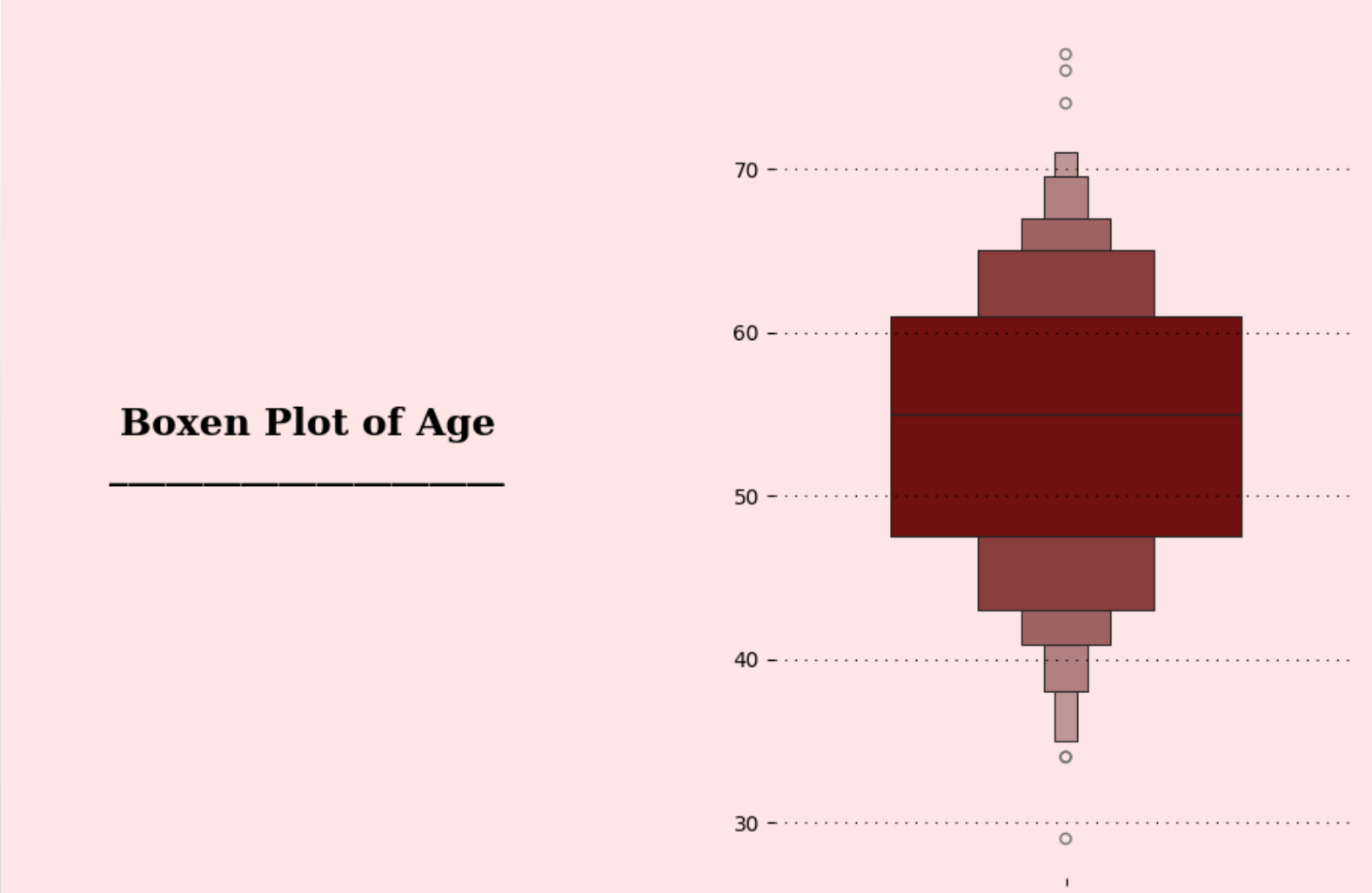
Descriptive Statistics – for Continuous Columns

	count	mean	std	min	25%	50%	75%	max
age	303.0	54.366337	9.082101	29.0	47.5	55.0	61.0	77.0
trtbps	303.0	131.623762	17.538143	94.0	120.0	130.0	140.0	200.0
chol	303.0	246.264026	51.830751	126.0	211.0	240.0	274.5	564.0
thalachh	303.0	149.646865	22.905161	71.0	133.5	153.0	166.0	202.0
oldpeak	303.0	1.039604	1.161075	0.0	0.0	0.8	1.6	6.2

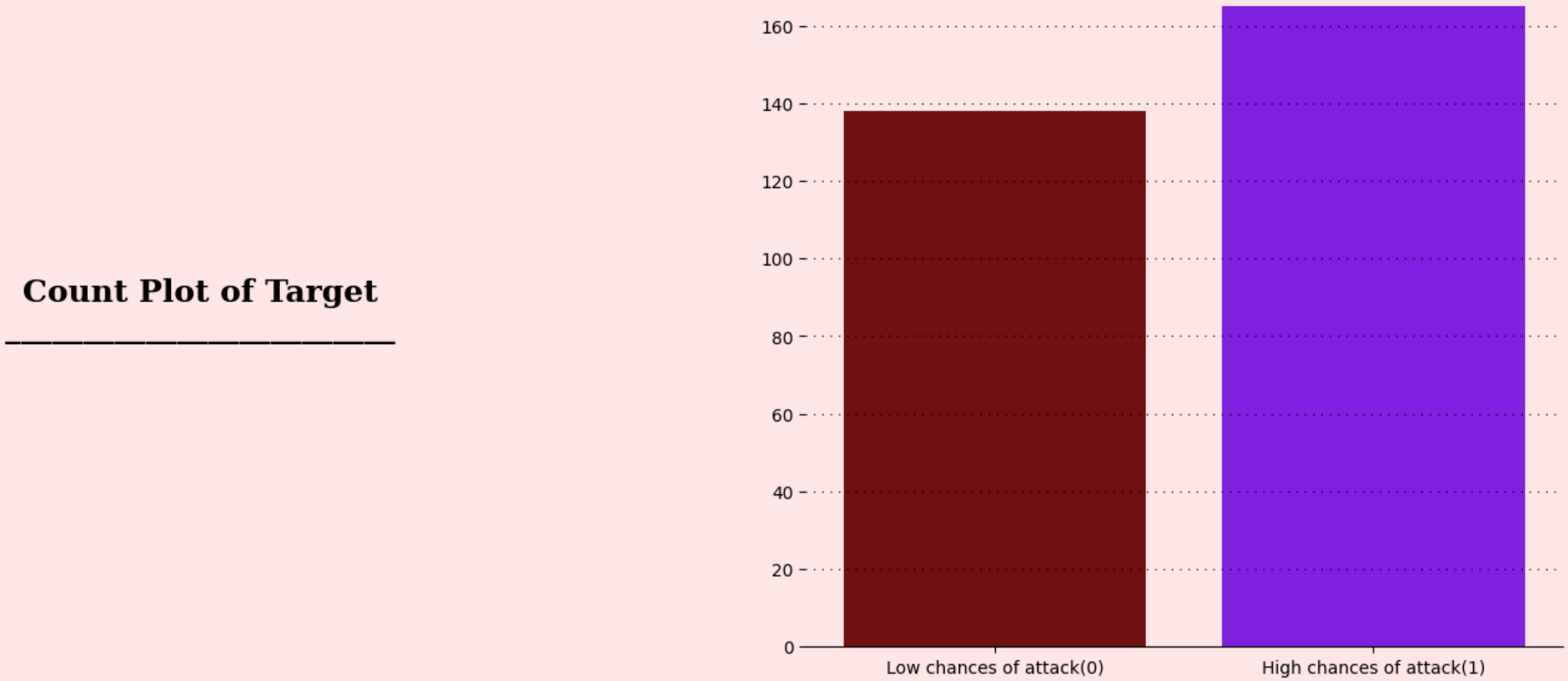
Count Plot of Sex



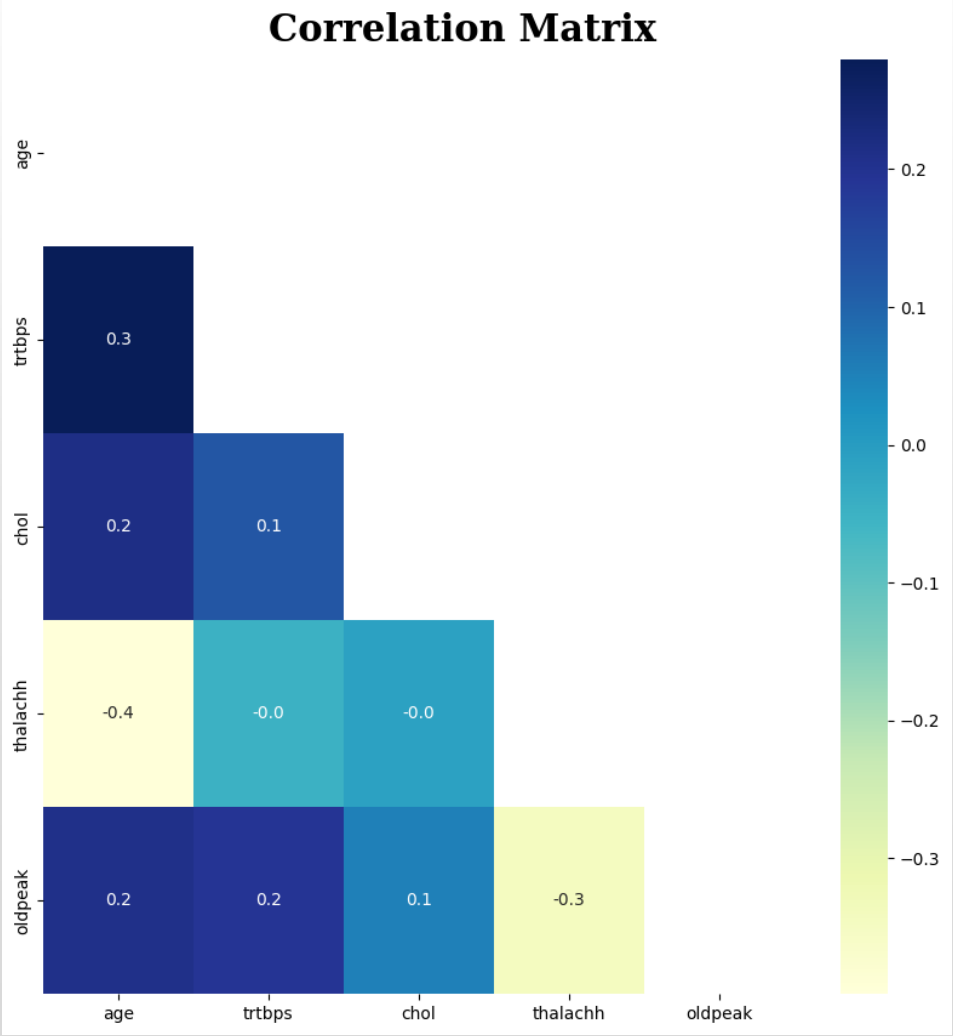
Boxen Plot of Age



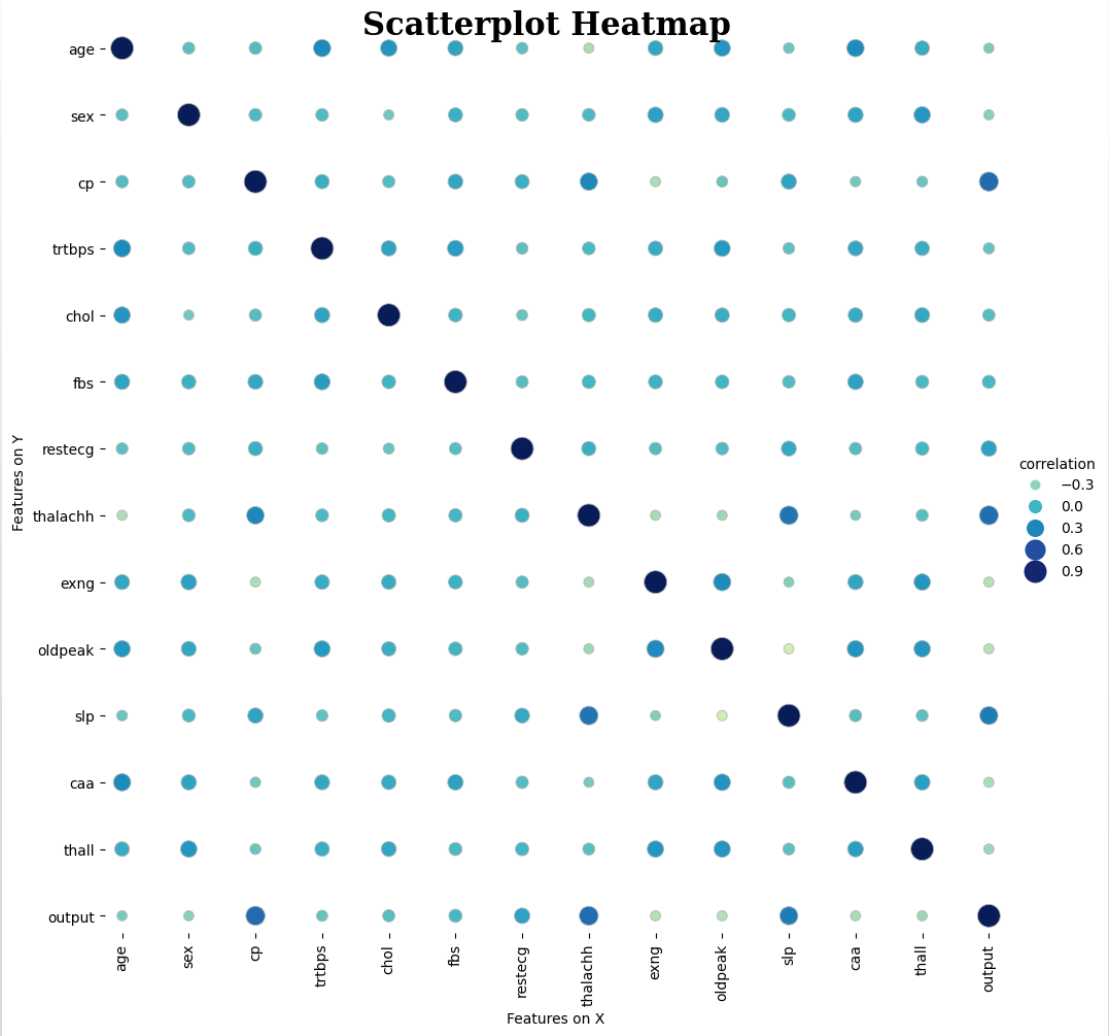
Count Plot of Target



Correlation between Continuous Features



Scatterplot Heatmap of Dataframe



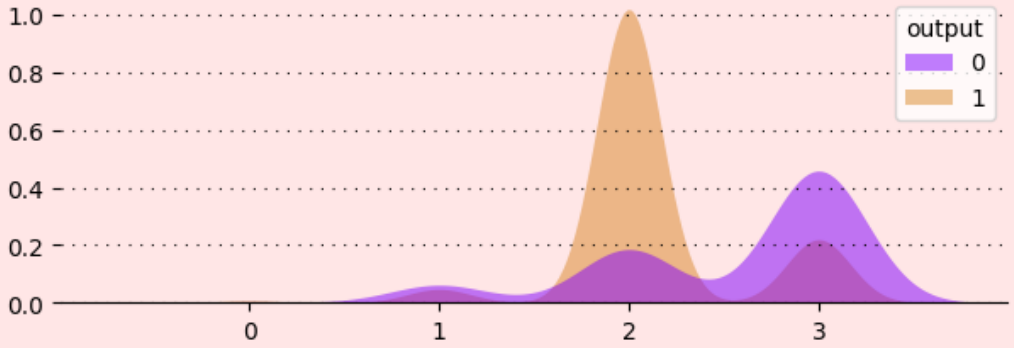
Intuitive Analyses

Heart Attack according to Sex

0 - Female
1 - Male

Distribution of Thall according to Target Variable

Thalium Stress Test Results
0, 1, 2, 3



Modeling and Accuracy

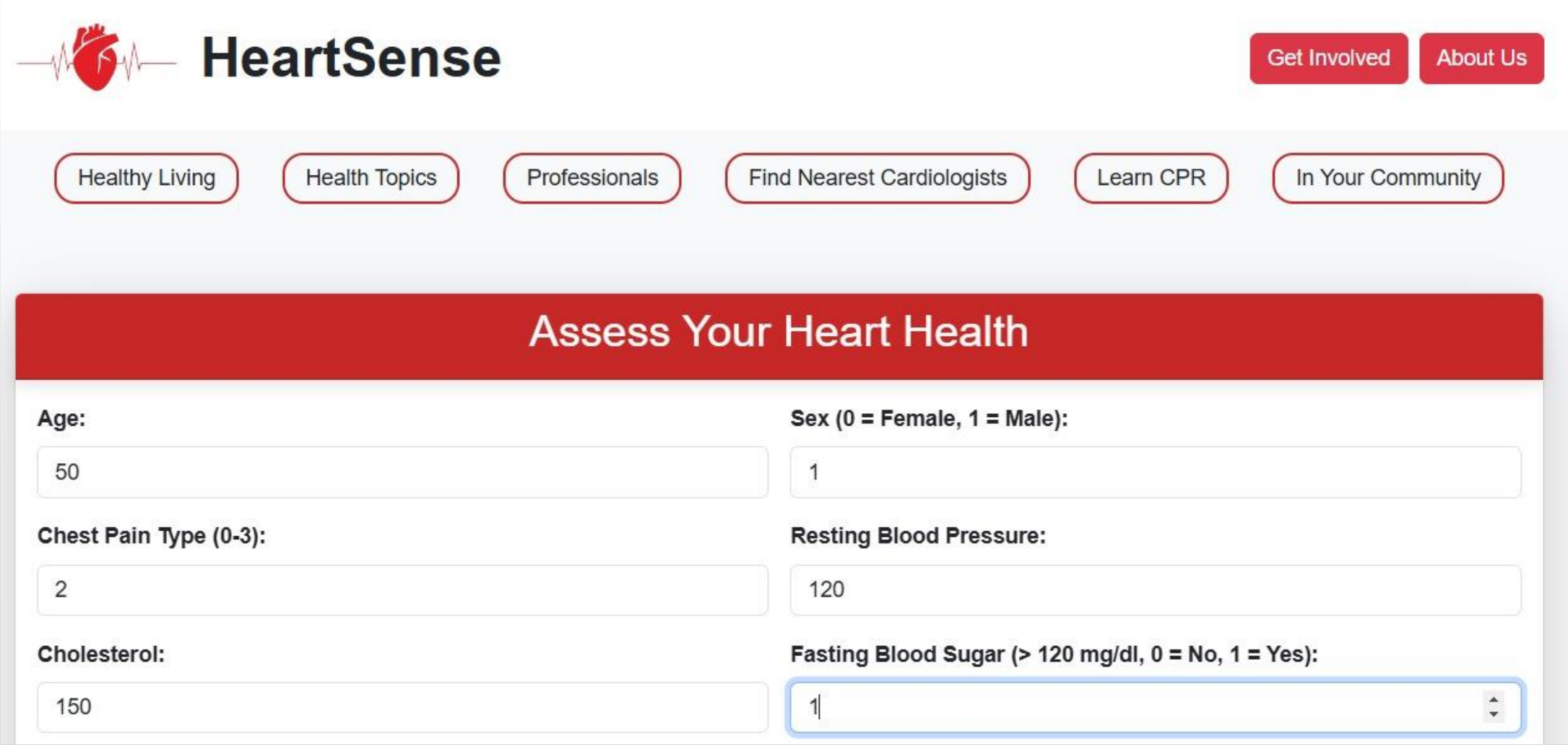
1) Linear Classifiers

SVM	0.8688
Hyper-Parameter Tuning of SVM	0.9016
Logistic Regression	0.9016

2) Prediction

Decision Tree	0.7868
Random Forest	0.7868
Gradient Boosting Classifier	0.8688

Making a Prediction



Making a Prediction

Exercise Induced Angina (0 = No, 1 = Yes):

1

ST Depression:

2.3

Slope of ST Segment (0-2):

2.3

Number of Major Vessels (0-3):

1

Thalassemia (0-3):

1

Predict

HeartSense Prediction Result:

Heart disease detected!

© 2025 Heart Health Inc. All Rights Reserved.

References

Dataset:

https://www.kaggle.com/code/namanmanchanda/heart-attack-eda-prediction-90-accuracy

EDA:

https://youtu.be/Liv6eeb1VfE?si=4LE7463YClR0wOW9

Linear Classifiers:

https://developer.dataiku.com/latest/getting-started/quickstart-tutorial/step2_ml_experiment.html

Tree Models:

https://blog.dataiku.com/tree-based-models-how-they-work-in-plain-english

Thank You!